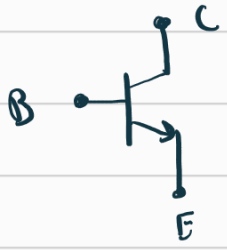
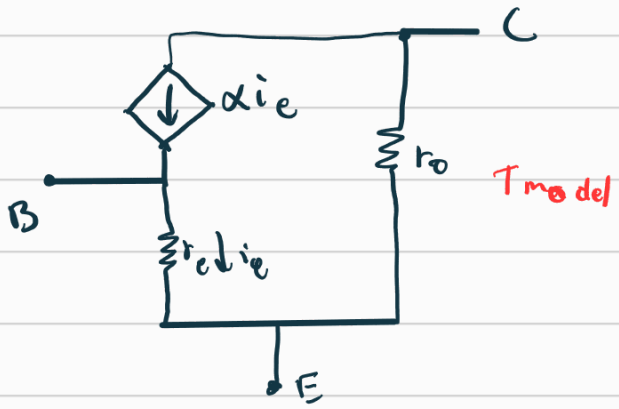
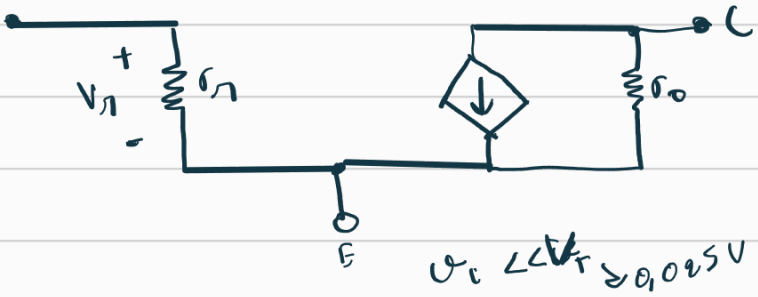
Small signal Models





T model

Π model



$v_i < V_T \gtrsim 0,025V$

Small signal model

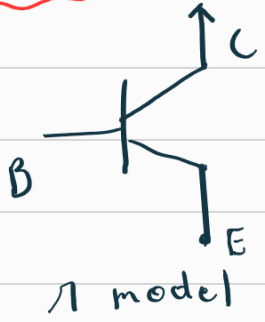
$r_o \rightarrow$ output resistance of BJT $= \frac{V_A}{I_C}$

$g_m \rightarrow$ transconductance $= \frac{I_C}{V_T}$

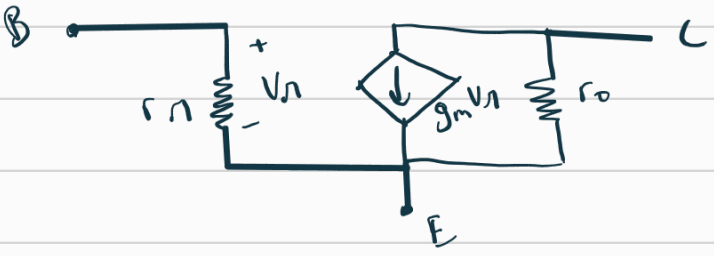
$r_e \rightarrow$ emitter resistance $= \frac{V_T}{I_E} \cong 1/g_m$

$r_\Pi \rightarrow$ base resistance $= (\beta+1).r_e$

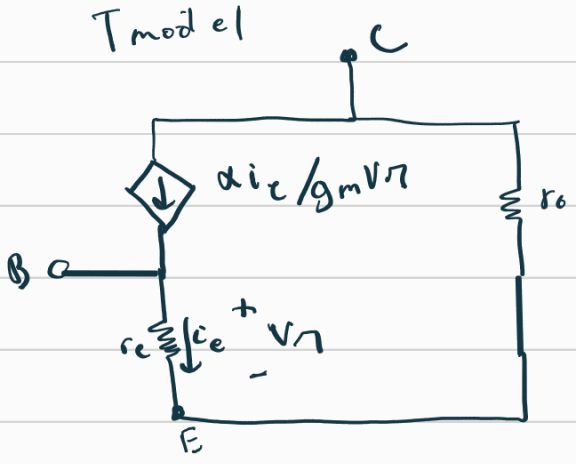
Practice

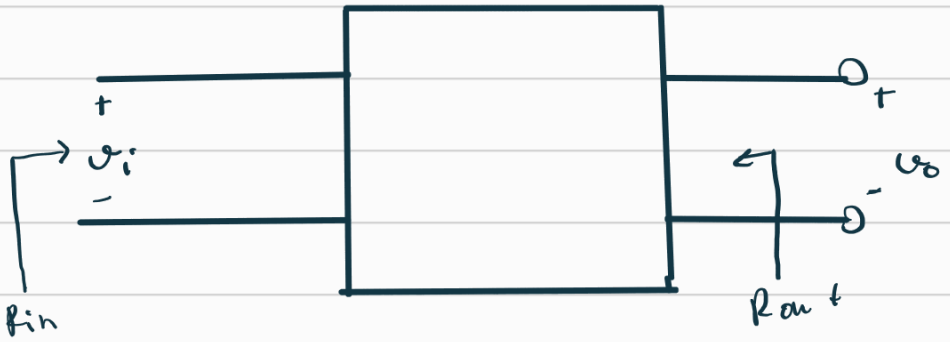


Π model



T model





Voltage Gain

$$A_v = \frac{v_o}{v_i}$$

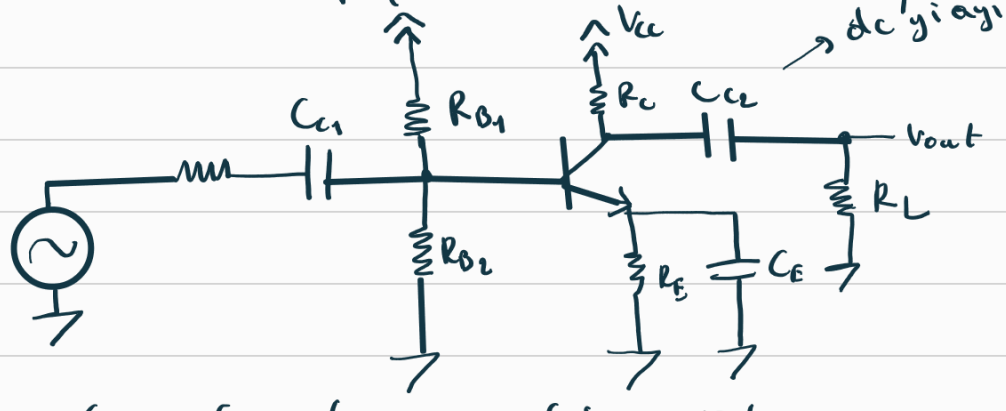
R_{in} → Input resistance

R_{out} → Output

Amplifier Topologies

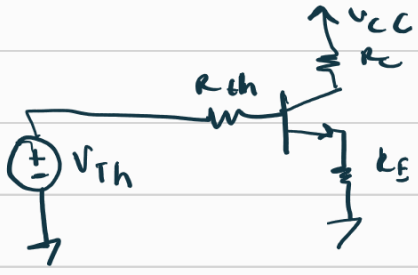
- Common Emitter
- Common Emitter with RE
- Common Base
- Common Collector (Emitter-Fdl)

Common Emitter amplifier



C_{c1}, C_{c2}, C_E are high values

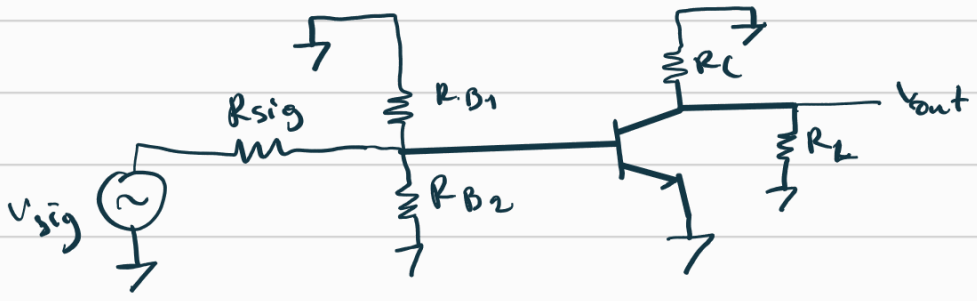
C are open circuit at Dc



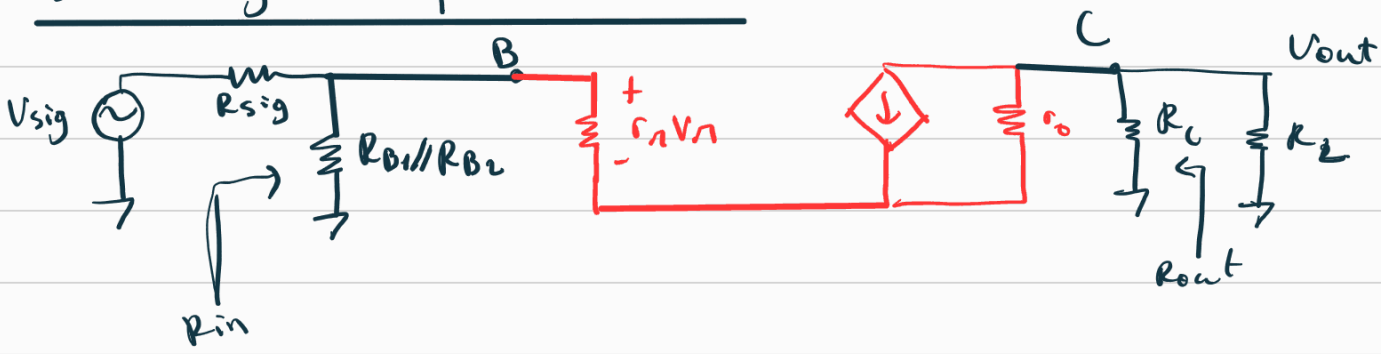
g_m, r_e, r_π, r_o

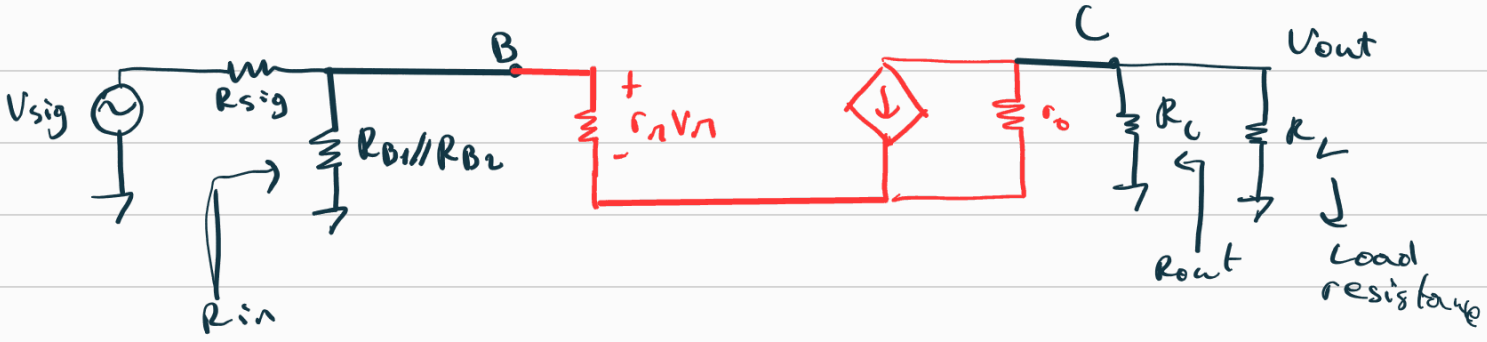
@ AC

- Now all caps are short circuit
- DC voltage sources are short circuit
- DC current sources are "open circuit"



Small signal equivalent circuit

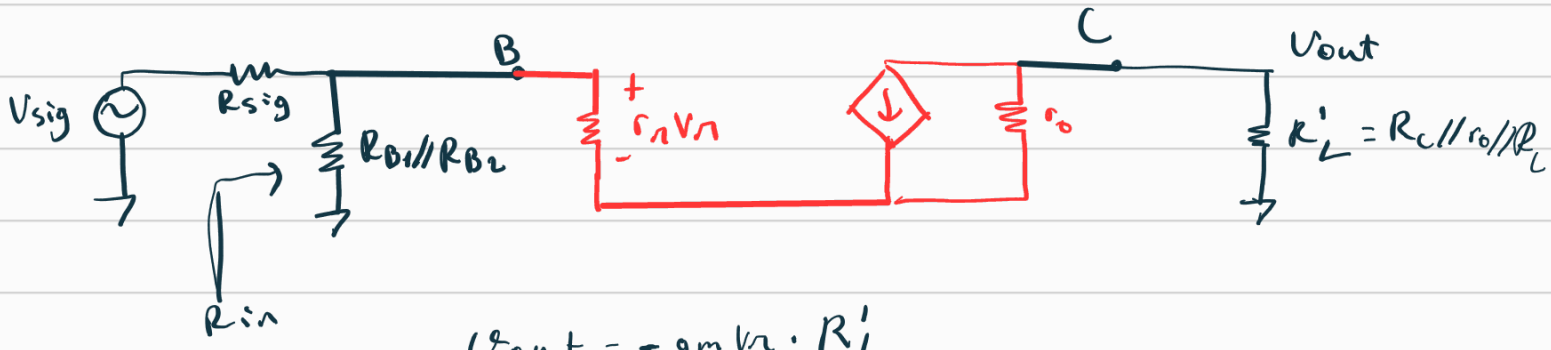




$A_V = \frac{v_{out}}{v_{in}}$, R_{in}, R_{out}

$R_{in} = R_{B1} // R_{B2} // r_\pi$

$R_{out} = R_C // r_o$



$$v_{out} = -g_m v_\pi \cdot R'_L$$

$$v_{in} = v_{sig}$$

$$v_\pi = \frac{v_{sig}}{R_{sig} + (R_{B1} // R_{B2} // r_\pi)} \times (R_{B1} // R_{B2} // r_\pi)$$

$$\frac{v_{out}}{v_{sig}} = -\left(\frac{R_{B1} // R_{B2} // r_\pi}{R_{sig} + (R_{B1} // R_2 // r_\pi)}\right) g_m \cdot R'_L$$

Due to large R_{si}

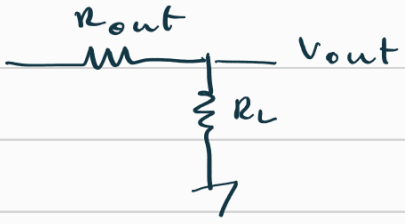
$$R_{sig} \approx 0 \Rightarrow = -g_m R'_L$$

→ High Voltage Gain $\cong -g_m R_L'$

→ Moderate Input Resistance → $R_{B1} // R_{B2} // r_\pi$

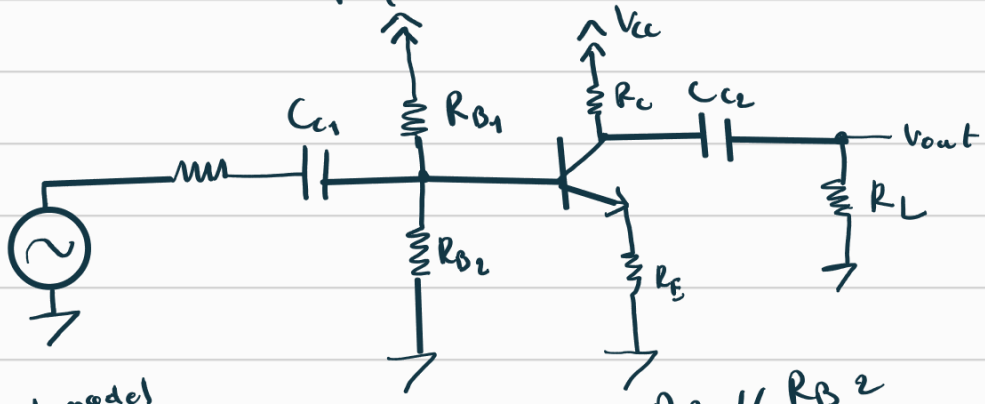
→ // High output Resistance → $R_C // r_o$

Rin büyük olsun isteriz



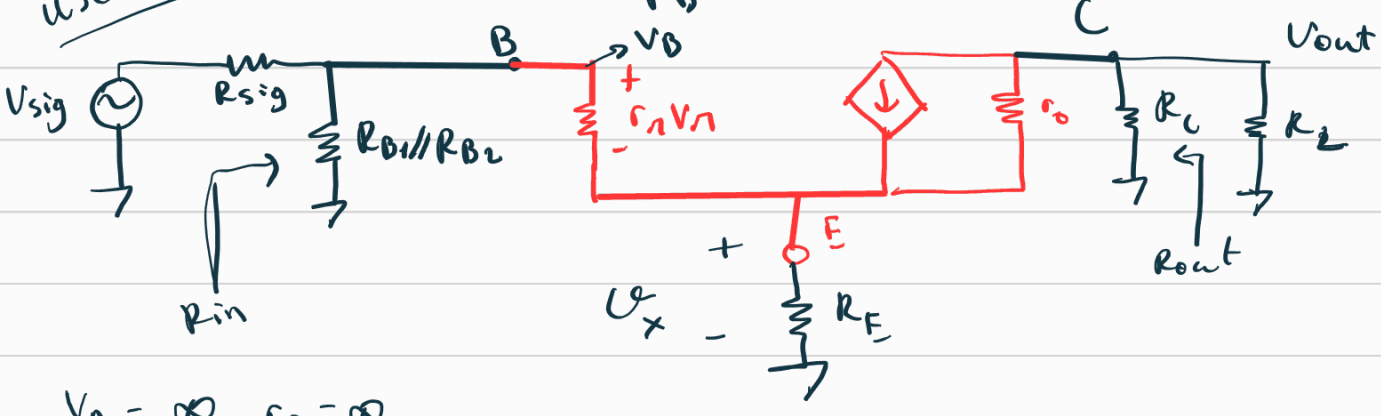
Rout küçük olmasını isteriz

Common Emitter amplifier with R_E



use t model

$R_B = R_{B1} // R_{B2}$



$V_A = \infty$, $r_o = \infty$

$$R_{in} = R_B \left[r_\pi + R_E (\beta+1) \right]$$

$$= R_B // \left[(\beta+1)(r_e + R_E) \right]$$

$$R_{out} \cong R_C$$

$$A_v = \frac{v_{out}}{v_{in}}$$

$$v_{out} = -g_m V_\pi (R_C // R_L) \qquad V_\pi \overset{?}{=} V_{sig}$$

$$v_x = \left(g_m V_\pi + \frac{V_\pi}{r_\pi} \right) R_E$$

$$= \left(\frac{V_\pi}{R_e} + \frac{V_\pi}{r_\pi} \right) R_E \cong \frac{v_\pi}{r_e} \cdot R_E$$

$$v_b = v_\pi \left(1 + \frac{R_e}{r_e} \right)$$

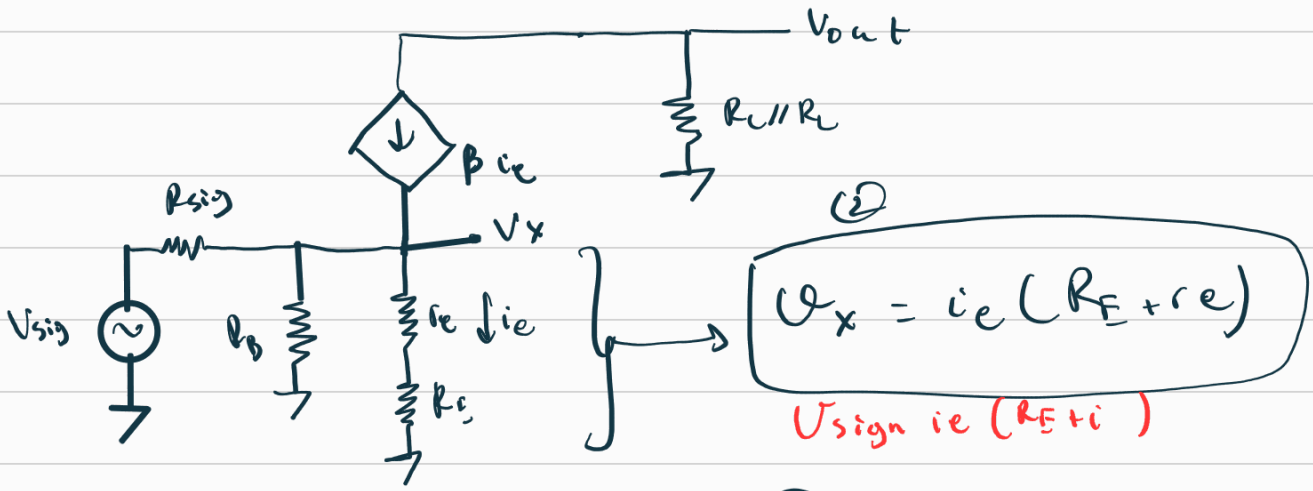
if $R_{sig} \approx 0$ $\quad v_b = v_{sig}$

$$\frac{v_{out}}{v_{in}} = \frac{-g_m \cancel{v_1}(R_C \| R_L)}{\cancel{v_1}\left(1+\frac{R_E}{r_e}\right)} \qquad r_e = \frac{1}{g_m}$$

$$\frac{v_{out}}{v_{in}} = \frac{-g_m (R_C \| R_L)}{r_e\left(1+\frac{R_E}{r_e}\right)}$$

$$\simeq -\frac{(R_C \| R_L)}{r_e + R_e}$$

$$A_v \simeq -\frac{R_C \| R_L}{R_E + r_e}$$



(2)

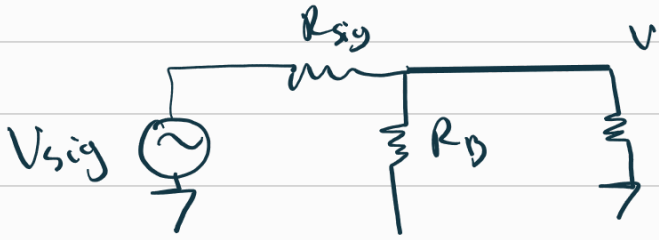
$$v_x = i_e(R_E + r_e)$$

$V_{sign}\ i_e\ (R_E + i$

(3)

$$v_{out} = -\alpha i_e (R_C \| R_L)$$

$$\frac{v_{out}}{V_{sig}} \qquad \frac{}{i_e(R_E + s_e)}$$

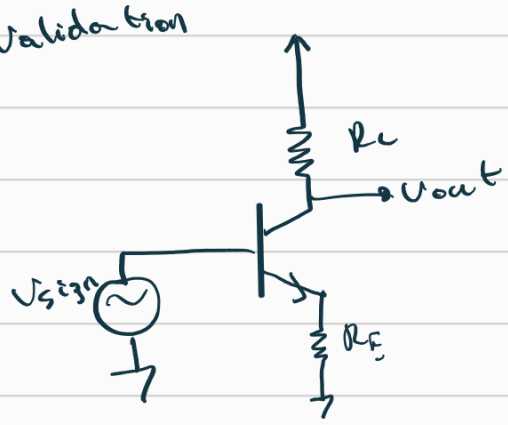


$$V_x = \frac{(\beta+1)(r_e+R_E) \| R_B \cdot V_{sig}}{R_{sig} + (\beta+1)(r_e+R_e) \| R_B}$$

R_T

$R_{signal} < R_T$

Validation



$$A_v = -\frac{R_c}{R_E}$$

- High input Imp
- Low voltage gain
- More output Imp
- More stable
- Independent of gm
- High bandwidth